

When Does Network Diversity Reflect Physical Diversity?

A Cross-Layer Audit of Internet Service Paths

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Abstract—Network-layer path diversity is often used as a proxy for the physical diversity supporting Internet services, but traceroute does not identify the submarine corridors that could carry an observed path. We present CLASP, a Cross-Layer Audit of Service Paths that constructs network-transition and feasible-corridor distributions from the same traceroute segments. We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements covering all 13 DNS Roots, three applications, and two dynamic multi-target topology references. The resulting audit contains 370 country–measurement units. Under projection-score weighting, the Spearman association between network and corridor concentration, which we call *Layer Agreement*, is 0.632 for 67 island and archipelagic service-facing units and 0.001 for 202 coastal- mainland units. The geographic difference remains in the same direction under equal-share and Top-1 candidate allocation, with differences of 0.461, 0.631, and 0.715. For the equal-share result, a country-cluster bootstrap gives a 95% interval of [-0.213, 0.902] for the 0.461 difference, and a country-label permutation gives $p = 0.161$; we therefore make no significance claim. The service-facing population is heterogeneous: 81 of 275 units have broad network support and concentrated corridor support, while 158 remain broad at both layers. These measurements show that the correspondence between network- layer and feasible physical-corridor diversity varies across the observed geographic settings; they do not establish a causal geographic mechanism or ground-truth cable use.

Index Terms—Internet measurement, submarine cables, traceroute, cross-layer analysis, anycast, path diversity

I. INTRODUCTION

Submarine cables form the physical backbone of the global Internet and carry nearly all intercontinental traffic [1]. Cable failures can isolate regions, degrade remote-service access, and redirect traffic toward longer or congested alternatives. Routes that share a cable or coastal corridor can be affected by the same physical event even when their network-layer paths appear different. Evaluating path diversity therefore requires measuring whether logically distinct routes also occupy distinct physical directions.

Path diversity is usually measured at the network layer. Traceroute records routers and Autonomous System (AS) transitions, so distinct hops or AS paths are counted as distinct alternatives. It does not identify the submarine infrastructure that could carry those paths. Two paths that appear independent in traceroute may leave a region through the same coast and rely on the same small set of feasible corridors. Whether network-layer diversity corresponds to physical-corridor diversity is therefore an empirical question.

Prior work supplies the cross-layer mappings needed to study this question. Internet Atlas, InterTubes, and iGDB connect observed Internet links to long- haul infrastructure; Nautilus associates IP links with candidate submarine cables; Calypso maps routes using cable layouts, network relationships, and inland connectivity; and Xaminer applies cross-layer maps to infrastructure risk analysis [2]–[7]. These systems identify physical infrastructure that could carry an individual link or path. We use such assignments to measure a population-level property: whether concentration among observed network transitions tracks concentration among feasible physical corridors.

We present *CLASP*, a Cross-Layer Audit of Service Paths. CLASP normalizes each traceroute into visible, geolocated hop-pair segments. Within every country–measurement population, ordered endpoint-AS transitions form the network distribution and feasible pairs of landing regions form the physical- corridor distribution. Both distributions contain the same candidate-bearing segments. Each segment contributes one unit of observation mass, allocated among candidate corridors by the existing projection score. We call the Spearman rank association between network and corridor concentration across a reported group of units *Layer Agreement*. The coefficient records an observed monotonic association; it is not a causal or predictive model.

Three questions organize the measurement. **RQ1** asks how frequently country–service paths contain a feasible inter-region submarine corridor. **RQ2** asks whether network-layer concentration tracks feasible-corridor concentration, and whether the association differs across geographic groups and candidate-allocation rules. **RQ3** asks how much variation is hidden by aggregate summaries: which populations contract, remain broad at both layers, remain concentrated, or expand after projection.

We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements. The corpus covers all 13 DNS Roots, Wikipedia, Reddit, Netflix Assets, and two dynamic multi-target measurements used as topology references. The aligned one-hour snapshot yields 3,291,063 atomic segments, of which 2,432,559 are mappable and 124,350 support at least one feasible inter-region corridor. After audit-quality filtering, the analysis contains 370 country– measurement units: 222 DNS Root, 53 application, and 95 topology-reference units. Each unit retains its source-country and measurement identity.

The main result is a geographic difference in the observed Layer Agreement. Under projection-score weighting, the coefficient is 0.632 among 67 island and archipelagic service-facing units drawn from 10 countries: units that are more concentrated at the network layer also tend to be more concentrated across feasible corridors. Among 202 coastal-mainland units drawn from 28 countries, the coefficient is 0.001, so the network-layer ordering contains essentially no information about the corridor ordering. The saved score-weighted artifact does not retain the per-unit table required for clustered inference. A country-clustered analysis of the available equal-share result gives a 95% bootstrap interval of $[-0.213, 0.902]$ for its 0.461 between-group difference and a two-sided country-label permutation value of $p = 0.161$. We therefore make no significance claim for the geographic contrast.

The geographic contrast remains in the same direction under all three candidate-allocation rules already computed. Equal-share allocation gives Layer Agreement of 0.380 for island and archipelagic units and -0.081 for coastal-mainland units; projection-score weighting gives 0.632 and 0.001; and Top-1 allocation gives 0.820 and 0.105. The corresponding between-group differences are 0.461, 0.631, and 0.715. The equal-share comparison contains 205 coastal-mainland units, whereas the saved score-weighted and Top-1 comparisons contain 202. The measured association increases as more observation mass is placed on higher-scoring candidates, but physical-route ground truth is unavailable, so these results do not identify one rule as the true assignment.

The service-facing population is also heterogeneous. Of 275 units, 81 (29.5%) have a broad network distribution and a concentrated corridor distribution, while 158 (57.5%) remain broad at both layers. Eighteen are concentrated at both layers and 18 expand from concentrated network support to broad corridor support. Singapore–Netflix illustrates continuous narrowing: its effective network-transition count of 19.86 becomes an effective corridor count of 3.41 without crossing the 80% concentration threshold. These outcomes show why a single contraction rate cannot represent all populations.

This paper makes three contributions. First, CLASP constructs paired network-transition and feasible-corridor distributions from identical traceroute segments, separating trace-level exposure from conditional concentration. Second, we measure a geographic difference in Layer Agreement that remains in the same direction under three existing candidate-allocation rules. Third, we provide a reproducible audit structure that records measurement IDs, configuration values, mapping states, and pipeline counts, allowing the reported distributions and their limitations to be inspected.

II. BACKGROUND AND RELATED WORK

A. Reading Service Paths at the Network Layer

Users reach services through domain names, CDN resources, replicas, and anycast targets. Routing selects the resulting path, and traceroute exposes the intervening IP hops

and AS transitions. These transitions are the usual unit of path-diversity measurement: they are counted, compared across vantage points, and used to describe how many distinct routes a service presents.

This is also the layer at which service reachability is commonly assessed. Prior work measures the user’s view of submarine criticality [8], characterizes anycast deployment and its network footprint [9], [10], and relates cable infrastructure to AS-level structure [11]. These measurements describe network-layer reachability. Our question is whether diversity at that layer corresponds to breadth across feasible physical directions.

The relationship is many-to-many. Different routers or AS transitions may project onto one landing-region corridor, while a recurring network transition may support several physical alternatives. A *landing-region corridor* is a direction-independent pair of bounded coastal regions connected by at least one feasible cable candidate. Exact landing stations preserve facility detail but split nearby access points serving the same direction; country-level inventories merge coastlines and international directions. Landing regions retain coastal structure between these two resolutions.

B. Mapping Paths onto Submarine Infrastructure

Event-oriented studies connect new submarine infrastructure to observed routing changes. Bischof et al. characterized Cuba’s connectivity before and after ALBA-1 and later formulated the task of connecting Internet observations to the worldwide cable mesh [12], [13]. Fanou et al. measured routing changes following new cable deployments [14]. Liu et al. combine resource discovery, RIPE Atlas traceroutes, geolocation, and propagation feasibility to quantify how users reach Web resources through submarine paths [8].

Cross-layer cartography supplies the assignments used by our measurement. Internet Atlas and InterTubes associate long-haul facilities with observed paths [2], [3]. iGDB integrates facilities, fiber, and network entities into a cross-layer map [4]. Nautilus associates IP links with candidate submarine cables and attaches confidence to each assignment [5]. Calypso combines cable layouts, network relationships, latency, and inland connectivity to map routes and defines Route Stress over those assignments [6]. Xaminer applies cross-layer maps to disaster-oriented infrastructure analysis [7]. Measurements of intercontinental links also identify layer-3 links and gateway routers [15].

These systems produce an assignment from a link or path to physical infrastructure that could plausibly carry it. Their validation evaluates the assignment using cable failures, targeted traceroutes, operator maps, or expert feedback. CLASP adopts this projection approach rather than claiming to identify ground-truth cable use. The individual assignments are inputs to our measurement; distributions aggregated over a path population are the objects we compare.

C. From Mapping to Distribution Measurement

An assignment does not by itself describe how observation mass is distributed across physical directions or whether that

distribution tracks the one visible at the network layer. We separate two quantities that are easy to conflate. *Submarine exposure* is the fraction of valid traceroutes containing a feasible inter-region corridor. *Corridor concentration* describes how candidate-bearing segment mass is distributed across corridors after entry. A country can enter the candidate space frequently and distribute observations widely, or enter infrequently with the exposed observations concentrated on one direction.

Country and measurement identity jointly define each observation population. Source geography and interconnection describe the context from which paths start; anycast, replica selection, and routing determine the destinations and paths observed during the measurement window. DNS Roots illustrate this interaction because functionally equivalent services operate independent anycast deployments. The three applications retain their own selected destinations. The two dynamic multi-target measurements deliberately sample a broader target population and serve only as topology references. They are not a matched control, and differences between families do not identify a service- deployment or target-selection effect.

CLASP aligns the statistical unit across layers. A network transition and its corridor support originate from the same atomic segment, so their concentration difference describes how a fixed observation population is organized. The framework next defines this paired construction, and Section IV records the aligned path and infrastructure inputs.

III. CLASP: THE PAIRED CROSS-LAYER AUDIT

A. Overview

CLASP constructs paired network-transition and feasible-corridor distributions for each country-measurement unit. It combines traceroute, probe metadata, IP geolocation and ASN mappings, AS context, and cable metadata. Figure 1 shows the processing order: path extraction creates atomic hop-pair segments, corridor projection constructs each segment's feasible landing-region corridor set, and aggregation forms the two distributions over the same candidate-bearing segments.

The construction keeps trace-level exposure separate from segment-level concentration. Exposure records whether a valid traceroute contains at least one feasible inter-region corridor. The paired distributions condition on candidate-bearing segments and describe how their observation mass is organized. Because these quantities use different denominators, CLASP reports them separately.

B. Path Extraction

We group traceroutes by probe country and measured service. This grouping retains the source-side connectivity context and the destination selection observed for each measurement. Hop countries remain segment attributes rather than grouping keys.

Each traceroute is normalized into its complete sequence of visible hops with locations, ASNs, and RTTs. Successive visible, geolocated hops form atomic segments. Each segment records endpoint locations, ASNs, countries, hop positions, RTT difference, and bridged-hop count. Repeated hops and

RTT samples retain their trace identity and ordering. When timeouts separate two visible hops, the bridged-hop count records the hidden gap for audit accounting.

When both endpoint ASNs are available, their ordered pair forms the segment's network-transition label. A segment missing one endpoint ASN receives an ordered country-pair fallback label, which is strictly coarser. We record the fallback share for each unit. Audit eligibility requires that share to be at most 30%, together with at least 30 candidate-bearing segments, 10 probes, and 3 probe ASNs. These thresholds remove units whose network labels or observation support are too sparse for the paired summaries; they are quality floors rather than guarantees of representativeness.

C. Feasible Landing-Region Corridors

a) *Landing stations and cable lifecycle.*: For each segment endpoint, CLASP retrieves landing stations within 50 km and pairs them through cable systems in service on July 1, 2026. The catchment allows endpoint locations to be associated with nearby coastal facilities at the resolution of the geolocation input. Explicit cable path or branch information defines direct landing-point adjacency. When the inventory does not provide that information, the pipeline retains reachability at the metadata's topology resolution and records that provenance state rather than presenting it as an observed cable traversal.

b) *Landing regions.*: Landing stations are grouped into regions with maximum pairwise distance 30 km. The diameter merges nearby facilities serving the same coastal direction while retaining separated coastal entry and exit patterns. A *landing-region corridor* is a direction-independent pair of regions connected by at least one feasible cable candidate. Several cable systems and exact landing pairs may support one corridor. Exact landing-pair, cable, and corridor identities remain linked in the outputs so that the aggregation can be audited at each resolution.

c) *Propagation feasibility.*: A candidate must satisfy geographic, lifecycle, and propagation constraints. Let D be the great-circle distance between its landing points, $v_f = 200$ km/ms the configured effective propagation speed in fiber, ΔRTT the segment's RTT difference, and $\tau = 5$ ms the RTT tolerance. The candidate is propagation-feasible when

$$\Delta RTT + \tau \geq \frac{2D}{v_f}. \quad (1)$$

The inequality asks whether the observed round-trip increment can accommodate the minimum round-trip propagation time between the landing points. When ΔRTT is non-positive or inconclusive, CLASP retains the geographic and lifecycle constraints and flags the segment. The 200 km/ms speed and 5 ms tolerance are fixed configuration values in the existing experiment set; Section VI treats the absence of a complete parameter sweep as a measurement limitation.

D. Candidate Allocation and Observation Mass

Duplicate cable rows belonging to the same corridor are collapsed within an atomic segment. For a segment s with

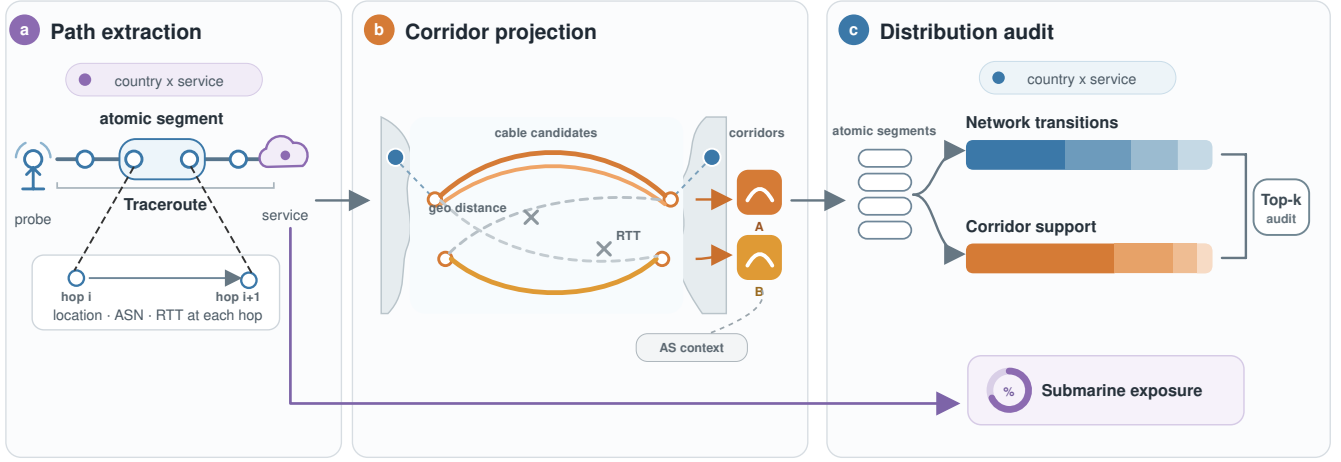


Fig. 1. CLASP extracts atomic segments, projects feasible landing-region corridors, and constructs paired network-transition and corridor distributions over the same candidate-bearing observations.

feasible corridor set $\mathcal{C}_s$, the primary analysis allocates its observation mass using the projection score:

$$w_{s,c} = \frac{\sigma_{s,c}}{\sum_{c' \in \mathcal{C}_s} \sigma_{s,c'}}, \quad \sum_{c \in \mathcal{C}_s} w_{s,c} = 1, \quad (2)$$

where $\sigma_{s,c}$ is the support assigned by the existing projection to corridor c , summed over its cable candidates. Every candidate-bearing segment contributes exactly one unit of total mass, so a segment with more candidates does not contribute more total influence.

Candidate ambiguity remains observable. We therefore retain two alternative allocations already computed in the artifact. Equal-share allocation assigns $w_{s,c} = 1/|\mathcal{C}_s|$. Top-1 allocation assigns the unit mass to the highest-scoring corridor. Equal sharing preserves every feasible direction but spreads mass without using the score; Top-1 removes that spread but discards lower-ranked candidates. Reporting all three rules shows which conclusions survive these treatments of ambiguity.

E. Mapping Resolution and Pipeline Accounting

Table I classifies each segment by input coverage and the size of its feasible corridor set. Single- and bounded multi-corridor segments form the physical distribution. Segments with no feasible corridor or insufficient inputs remain in pipeline accounting, and every valid traceroute remains in the exposure denominator.

Candidate-set size and aggregate concentration describe different levels. A unit may contain many bounded multi-corridor segments yet concentrate after their allocated mass repeatedly supports the same corridors. CLASP therefore retains the resolution state instead of using candidate multiplicity as a substitute for distribution breadth.

F. Paired Distributions

For an auditable country–measurement unit u , let S_u denote its candidate-bearing atomic segments and $\ell(s)$ the ordered

TABLE I
SEGMENT-LEVEL PHYSICAL MAPPING RESOLUTION.

State	Operational definition
Single corridor	Feasibility inputs available and $ \mathcal{C}_s = 1$
Bounded multi-corridor	Feasibility inputs available and $ \mathcal{C}_s > 1$
No feasible corridor	Feasibility inputs available and $ \mathcal{C}_s = 0$
Insufficiently resolved	Insufficient location, path-visibility, or meta-data fields for stable corridor construction

network label of segment s . The network-transition probability of label t is

$$p_u^N(t) = \frac{\sum_{s \in S_u} \mathbf{1}[\ell(s) = t]}{|S_u|}. \quad (3)$$

The corridor distribution sums the masses in Eq. (2) for each corridor and normalizes within the same S_u . Before normalization, both distributions have total mass $|S_u|$. Their difference therefore describes how the same candidate-bearing observations are organized at the two layers, not a comparison between different trace populations. Direction is preserved for network transitions; the physical corridor is direction-independent because it represents a coastal connection rather than forwarding order.

Submarine exposure uses the complete valid-trace population. A traceroute is exposed when at least one of its segments has a feasible inter-region corridor. The number of exposed traces divided by all valid traces in the unit gives the exposure rate. This denominator differs intentionally from $|S_u|$, which conditions the distributions on candidate-bearing segments.

G. Distribution Summaries and Layer Agreement

Three metrics summarize each distribution. Top-2 share is the sum of the two largest probabilities and records mass in the dominant categories. Effective category count is

TABLE II
CLASP MEASUREMENT CONFIGURATION.

Parameter	Value
Endpoint landing catchment	50 km
Landing-region maximum diameter	30 km
Fiber propagation speed	200 km/ms
RTT tolerance	5 ms
Same-city threshold	25 km
Cable lifecycle date	July 1, 2026
Inconclusive RTT	Retained and flagged
Timeout-bridged segment	Retained and flagged
Path scope	Complete visible sequence

$\exp[-\sum_i p_i \log p_i]$ and reports breadth on the scale of an equally weighted category count. Normalized entropy divides Shannon entropy by the logarithm of observed support size and measures evenness relative to that support. Network-to-corridor differences are computed within each unit.

Layer Agreement is the Spearman rank association, across a reported set of units, between network Top-2 share and corridor Top-2 share. A positive coefficient means that units ordered as more concentrated at the network layer tend to be ordered as more concentrated at the corridor layer. It describes an observed monotonic association and is not a causal or predictive model. We report it separately by measurement family, geography, and candidate- allocation rule.

For the frozen equal-share result, we quantify geographic uncertainty at the level where geography is assigned. A cluster-bootstrap replicate samples countries with replacement within each geographic group and carries every country-measurement unit from each sampled country together. We use 50,000 replicates and report percentile intervals. A separate two-sided reference test permutes the geographic labels of whole countries 100,000 times while retaining the observed number of countries in each group. Both procedures use seed 20260919. The saved projection-score and Top-1 artifacts do not retain the per-unit summaries needed to repeat this clustered analysis, so their coefficients remain descriptive and we do not make significance claims.

A common 80% Top-2 threshold supplies four descriptive classes: broad at both layers, concentrated at both, network-broad/corridor-concentrated, and network-concentrated/corridor-broad. Continuous Top-2 shifts, effective counts, and normalized entropy accompany the classes so that threshold crossings do not replace the underlying distributions.

IV. DATASETS

All inputs are aligned to July 1, 2026 (UTC). Network-path observations, IP and AS annotations, and cable lifecycle records therefore describe a common analysis period. We combine these observations with network and submarine-cable metadata.

A. RIPE Atlas Measurements

Our network observations comprise 18 public RIPE Atlas IPv4 traceroute measurements collected from 00:00 to

01:00 UTC. Thirteen measurements cover independently operated anycast DNS Root services. Wikipedia, Reddit, and `assets.nflxext.com` provide three application-facing observations. Measurements 5051 and 5151 use dynamic targets as multi-target UDP and ICMP topology references. Table III summarizes the corpus after canonical trace-identity deduplication.

Each record retains the destination address returned by RIPE Atlas. This preserves the multi-target structure of the two dynamic-target measurements and the destination instances selected by service measurements during the aligned window.

The measurement families do not form a controlled comparison. The two dynamic- target measurements sample many destinations during the window, whereas each service measurement retains the destinations selected for that service. We use the former as topology references and describe differences between the observed populations; we do not attribute those differences to service deployment or target selection.

B. Supporting Datasets

Table IV lists the supporting datasets and versions. RIPE Atlas probe metadata supplies the probe country and source-network context [16]. IPinfo Location and ASN databases annotate visible hops with geographic and AS information [17].

The Submarine Cable Map provides the cable and landing-point inventory, including lifecycle and organizational fields [1]. The analysis includes systems in service on the measurement date. CAIDA AS Relationships and the derived owner-AS associations supply supplementary ranking context [18]. A versioned country taxonomy classifies probe countries as island or archipelagic, landlocked, or other coastal.

V. RESULTS

The results first record how often paths enter the feasible inter-region candidate space, then compare network and corridor concentration within the same candidate-bearing observations. Layer Agreement is the primary cross- layer summary. Absolute exposure and breadth remain separate results because they use different denominators and answer different questions.

The processing pipeline retains 490,911 valid traceroutes from 707,314 raw records and yields 3,291,063 atomic segments. Of 2,432,559 mappable segments, 555,119 reach candidate landing points, 171,232 satisfy the corridor constraints, and 124,350 support at least one feasible inter-region corridor. The equal-share audit records 71,990 single-corridor and 52,360 bounded multi- corridor segments. After the eligibility filters in Section III-B, the analysis contains 370 auditable country-measurement units: 222 DNS Root, 53 application, and 95 topology- reference units.

A. Inter-Region Candidate Exposure

Inter-region candidates occur infrequently for most service-facing populations. The DNS family contains 945 country-measurement units with at least 30 valid traceroutes across 77 countries. Pooling eligible Root observations within each

TABLE III
RIPE ATLAS MEASUREMENT CORPUS.

Group	Target	MSM ID(s)	Raw	Valid
DNS Roots	A-M Root	5009, 5010, 5011, 5012, 5013, 5004, 5014, 5015, 5005, 5016, 5001, 5008, 5006	534,276	317,873
Applications	Wikipedia; Reddit; Netflix Assets	86710103; 176906957; 176517335	115,596	115,596
Topology references	IPv4 UDP; IPv4 ICMP	5051; 5151	57,442	57,442
Total	18 measurements	-	707,314	490,911

TABLE IV
SUPPORTING DATASETS.

Dataset	Version	Content
RIPE Atlas probe metadata	2026-07-01	Probe country and source ASN
IPinfo Location MMDB	2026-07-01	Hop geolocation
IPinfo ASN MMDB	2026-07-01	IP-AS mapping
Submarine Cable Map	2026-07-01	694 cables; 1,916 landing points
CAIDA AS Relationships	2026-07-01 release	AS relationships
Owner-AS mapping	2026-07-01 inputs	Owner-AS associations
Country geography taxonomy	2026-07-01	Operational geography classes

country by valid-trace count yields median DNS exposure of 5.50% (IQR 1.10–23.04%; mean 17.80%). The difference between the median and mean records a pronounced upper tail across countries.

The three application measurements occupy the same low-exposure range. Median country–measurement exposure is 1.90% for Wikipedia, 1.96% for Reddit, and 1.43% for Netflix Assets. The dynamic multi-target topology references reach 23.24% for MSM 5051 and 23.81% for MSM 5151. These figures describe how often paths in each observed population contain a feasible inter-region corridor; they do not describe concentration after entry.

Country-matched comparisons retain source geography while changing the target population. DNS exposure is lower than MSM 5051 in 70 of 77 shared countries, with a median paired difference of -14.52 percentage points. It is lower than MSM 5151 in 54 of 59 shared countries, with a median difference of -14.82 points. The paired direction holds for 90.9% and 91.5% of shared countries, respectively. Because the target populations differ by design, the comparison is descriptive and does not identify a deployment or target-selection effect.

B. Layer Agreement Differs by Geography

The association between network and corridor concentration differs across the observed service-facing geographic groups. Under projection-score weighting, Layer Agreement is 0.632 across 67 island and archipelagic units and 0.001 across 202 coastal-mainland units. These units represent 10 and 28 countries, respectively, and the observed coefficient difference is 0.631. Landlocked units are not included in either group. The coefficients are descriptive associations; the saved score-weighted per-unit table required for a country- clustered uncertainty interval is not available.

The geographic contrast remains in the same direction under all three candidate-allocation rules. As allocation moves from

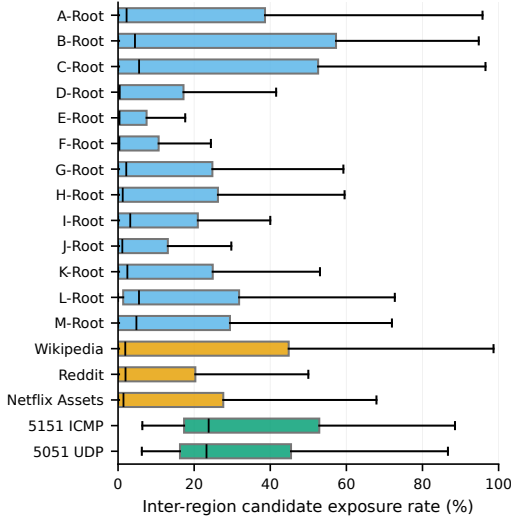
TABLE V
SERVICE-FACING LAYER AGREEMENT. PARENTHESES GIVE UNIT COUNTS.

Allocation	Island/arch.	Coastal mainland	Difference
Equal share	0.380 (67)	-0.081 (205)	0.461
Projection score	0.632 (67)	0.001 (202)	0.631
Top-1	0.820 (67)	0.105 (202)	0.715

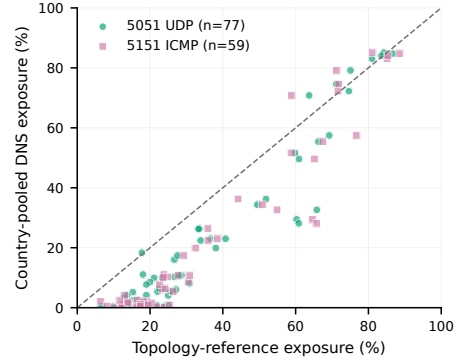
equal shares to projection-score weighting and then Top-1, the island coefficient increases from 0.380 to 0.632 and 0.820, while the coastal-mainland coefficient changes from -0.081 to 0.001 and 0.105. The between-group difference consequently increases from 0.461 to 0.631 and 0.715. The equal-share result contains three additional coastal-mainland DNS units that are absent from the saved score- weighted and Top-1 comparison. The monotonic sequence therefore describes the available allocation-specific cohorts rather than a fully paired estimator comparison. It shows that the geographic ordering does not arise under only one allocation rule, but it does not isolate allocation as the cause of the coefficient change. Without physical-route ground truth, the comparison also does not establish that one rule recovers the true association.

The reproducible equal-share result supports a country-level uncertainty check. Its 67 island and archipelagic units represent 10 countries, and its 205 coastal-mainland units represent 30 countries. Country-cluster bootstrap intervals are [-0.203, 0.760] for the island coefficient, [-0.284, 0.238] for the coastal-mainland coefficient, and [-0.213, 0.902] for their 0.461 difference. A two-sided country-label permutation gives $p = 0.161$. The wide interval for the difference includes zero, so the repeated direction across allocation rules is reported as a descriptive robustness pattern rather than as statistically established geographic separation.

The result does not imply that geography causes the two layers to agree or disagree. It records how the two concentration



(a) Country-measurement exposure.



(b) Country-matched comparison.

Fig. 2. Inter-region candidate exposure in the observed measurement populations.

rankings relate within the measured snapshot. The existing results constrain possible explanations but do not identify a causal mechanism.

C. Absolute Exposure and Breadth by Geography

The geographic groups also differ in how often DNS paths enter the candidate space. Giving each country equal weight, island and archipelagic countries have median country-pooled DNS exposure of 49.61% across 9 countries, compared with 7.88% for 54 other coastal countries and 1.70% for 14 landlocked countries. This ordering concerns exposure, not Layer Agreement.

Conditional equal-share corridor breadth follows a different ordering. Island and archipelagic countries have median corridor Top-2 share of 53.3% and 5.58 effective corridors. Other coastal countries reach 75.4% and 3.39. The two represented landlocked countries reach 80.2% and 2.51. Across 38 countries with both pooled exposure and an auditable corridor summary, exposure has Spearman association -0.29 with corridor Top-2 share and 0.30 with effective corridor count. These quantities show that frequent entry and conditional concentration are not interchangeable.

D. Layer Agreement within Measurement Families

Family-level coefficients do not reproduce the geographic separation. Table VI reports Layer Agreement within each observed measurement family. Under projection-score weighting, the coefficient is 0.176 for 219 DNS units, 0.229 for 53 application units, and 0.012 for 95 topology-reference units. Equal-share and Top-1 allocation change all three coefficients, with the largest change occurring for DNS Roots. The existing artifacts contain 222 DNS units in the equal-share audit and 219 in the family allocation comparison; the status of the remaining three must be reconciled before a final artifact release.

TABLE VI
LAYER AGREEMENT BY MEASUREMENT FAMILY.

Family	n	Equal	Score	Top-1
DNS Roots	219	0.096	0.176	0.330
Applications	53	0.030	0.229	0.372
Topology references	95	0.139	0.012	0.122

Absolute equal-share corridor breadth also differs among the observed families. Table VII reports network and corridor medians. DNS Roots and applications have median corridor Top-2 shares of 72.0% and 70.3%, compared with 30.6% for topology references. Their median effective corridor counts are 3.84 and 3.51, compared with 18.01. These are summaries of different target populations, not a controlled estimate of a service effect.

The high-concentration tail follows the same descriptive family ordering. The leading two corridors receive at least 80% of mass in 83 of 222 DNS units (37.4%), 16 of 53 application units (30.2%), and 4 of 95 topology-reference units (4.2%). Wikipedia, Reddit, and Netflix Assets have median corridor Top-2 shares of 68.3%, 71.0%, and 71.6%, with 4.25, 3.25, and 3.47 effective corridors. Their median Top-2 shifts are 11.0, 25.6, and 23.6 percentage points, respectively.

Normalized entropy supplies a support-relative view. Median entropy reduction is 0.126 for DNS Roots, 0.164 for applications, and 0.081 for topology references. The family summaries therefore agree on observed absolute breadth and concentration, while the Layer Agreement coefficients show that within-family rankings remain weak and allocation-sensitive.

E. Cross-Layer Outcomes Are Heterogeneous

The 80% Top-2 threshold identifies several service-facing outcomes rather than a universal contraction. Among 275

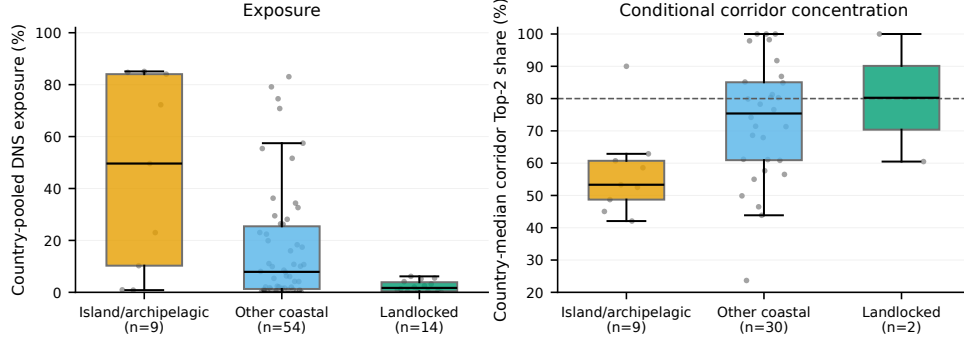


Fig. 3. DNS exposure and equal-share corridor concentration by geography. Country counts refer to the DNS country-level summary and are distinct from the service-facing unit counts in Table V.

TABLE VII
EQUAL-SHARE NETWORK-TRANSITION AND CORRIDOR DISTRIBUTIONS.

Family	Units	Net. Top-2	Corr. Top-2	Δ Top-2	Eff. network	Eff. corridor
DNS Roots	222	53.1%	72.0%	+11.0 pp	7.45	3.84
Applications	53	45.2%	70.3%	+21.0 pp	9.89	3.51
Topology references	95	27.3%	30.6%	+2.7 pp	24.34	18.01

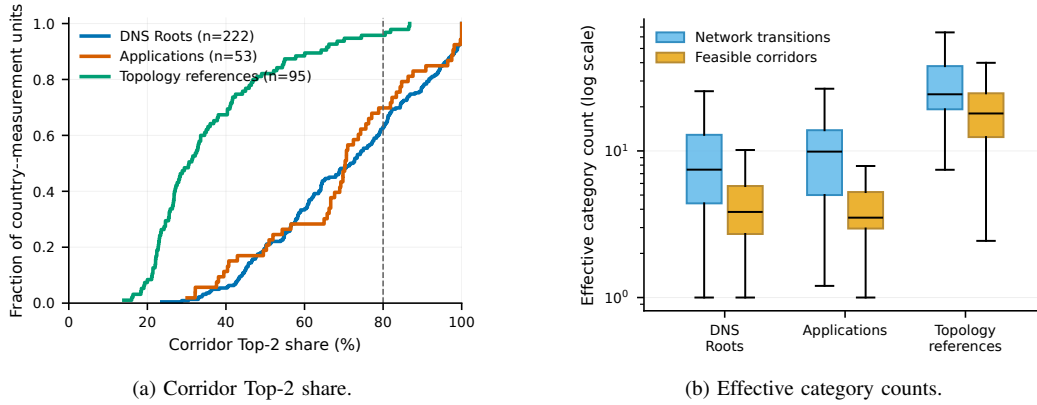


Fig. 4. Equal-share corridor concentration and breadth by measurement family. The families contain different target populations.

service-facing units, 81 (29.5%) are network-broad/corridor-concentrated, 158 (57.5%) remain broad at both layers, 18 (6.5%) are concentrated at both, and 18 (6.5%) are network-concentrated/ corridor-broad. Four of 95 topology-reference units (4.2%) enter the first class; the other 91 remain broad at both layers.

DNS Roots and applications enter the network-broad/corridor-concentrated class at similar rates: 66 of 222 DNS units (29.7%) and 15 of 53 application units (28.3%). Their median continuous Top-2 shifts are 11.0 and 21.0 percentage points. Across all service-facing units, corridor Top-2 share increases in 184 of 275 (66.9%), and effective category count contracts in 204 (74.2%). Of 239 units with a broad network distribution, 178 (74.5%) move upward in Top-2 share: 81 cross the threshold, while 97 remain broad at both layers with a median increase of 6.3 percentage points.

Three existing cases distinguish continuous narrowing, ex-

pansion, and changes in absolute breadth from support-relative evenness. **Singapore–Netflix** moves from 32.5% network Top-2 share to 65.0% corridor Top-2 share, while its effective count contracts from 19.86 to 3.41. It narrows without crossing the 80% threshold. **New Zealand–Netflix** moves from 63.6% to 37.6%, with effective count expanding from 5.18 to 6.92. **Singapore–H-Root** moves from 39.6% to 50.1%, and its effective count contracts from 20.67 to 5.99 while normalized entropy changes only slightly.

Bounded multi-corridor segments account for 99.9%, 78.1%, and 98.2% of candidate-bearing observations in these three cases. Repeated overlap among the bounded sets produces different aggregate distributions under the same projection and aggregation rules. The cases and four-class counts show why neither a universal contraction claim nor a single retention ratio represents the measured service-facing population.

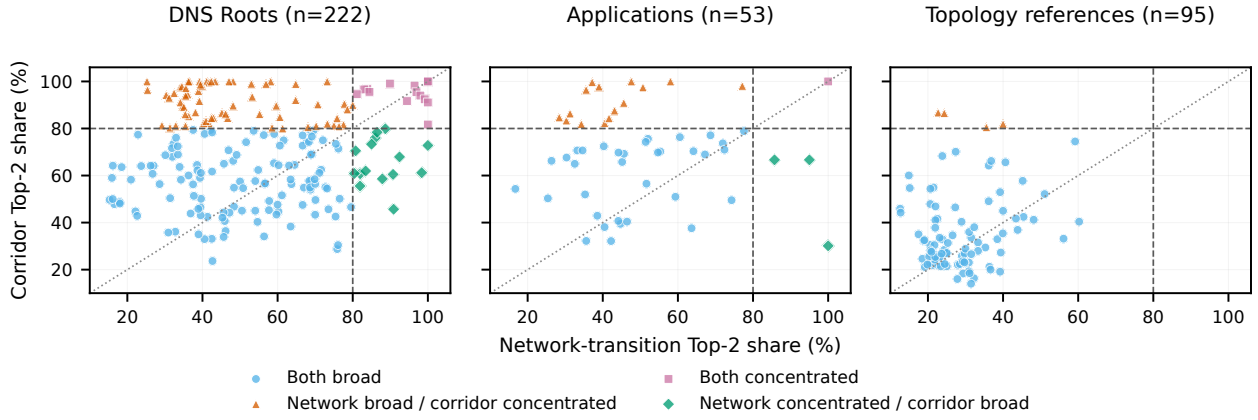


Fig. 5. Equal-share network-transition and corridor Top-2 shares for 370 auditable units. The 80% lines define the descriptive four-class view; the diagonal separates increases from decreases in concentration.

VI. ROBUSTNESS, SCOPE, AND IMPLICATIONS

The cross-layer distributions are feasible-corridor estimates rather than observations of cable use. This section reports the robustness checks already present in the experiment set and states which uncertainties remain unmeasured. It then separates the measured quantities from their operational interpretation.

A. Candidate-Allocation Robustness

Candidate allocation changes the magnitude of Layer Agreement but not the geographic ordering in the existing results. As Table V reports, the island-coastal coefficient difference is 0.461 under equal shares, 0.631 under projection-score weighting, and 0.715 under Top-1 allocation. The corresponding island coefficients are 0.380, 0.632, and 0.820; the coastal-mainland coefficients are -0.081, 0.001, and 0.105.

The allocation results do not all use an identical coastal-mainland cohort. Equal sharing retains 205 coastal-mainland units; the saved projection-score and Top-1 comparison retains 202. For equal sharing, country-cluster 95% intervals are [-0.203, 0.760] for island and archipelagic units and [-0.284, 0.238] for coastal-mainland units. The interval for their 0.461 difference is [-0.213, 0.902], and the two-sided country-label permutation value is $p = 0.161$. These results quantify substantial country-level uncertainty and do not establish statistical separation. The current artifact cannot supply the corresponding clustered intervals for the 0.632 and 0.001 score-weighted coefficients because its per-unit score-weighted table was not retained.

This comparison addresses one source of ambiguity directly. Equal allocation spreads each segment's mass across every feasible corridor, projection-score weighting places more mass on better-supported candidates, and Top-1 retains only the highest-scoring candidate. The increasing coefficients show that the estimated agreement depends on candidate allocation. The repeated geographic ordering shows that the observed contrast is not produced by one of these three rules. The experiment does not provide cable-use ground truth and therefore

TABLE VIII
A-ROOT LANDING-REGION DIAMETER SENSITIVITY.

Diameter	Single-corridor	Median Top-2	Units
10 km	47.4%	53.6%	16
20 km	46.2%	62.8%	15
30 km	54.8%	64.4%	15
40 km	90.1%	81.6%	13
50 km	94.8%	84.9%	13

does not establish which coefficient is closest to the physical route actually used.

B. Landing-Region Resolution

The available landing-region-diameter analysis covers the A-Root subset, not all 18 measurements. It varies maximum diameter over 10, 20, 30, 40, and 50 km while holding the endpoint catchment at 50 km and the RTT tolerance at 5 ms. Table VIII reports the recorded values. Exact landing-pair and cable candidate sets have mean Jaccard 1.0 on segments shared with the 30-km baseline, but the candidate-bearing population and number of auditable units change with diameter. The sensitivity therefore cannot be described as isolating grouping resolution over a fixed global cohort.

The 10–30-km settings retain a finer coastal partition than the 40–50-km settings. Single-corridor share rises to 90.1% and 94.8% at 40 and 50 km, and median Top-2 share reaches 81.6% and 84.9%. Because the analysis is limited to A-Root, it does not establish that the headline geographic difference is invariant to landing-region diameter. No complete existing sweep recomputes the main result across endpoint catchment, propagation speed, or RTT tolerance. We therefore use 30 km as an operational reporting resolution, not as an estimated optimum or a globally validated physical scale. Within the A-Root check, 10–30 km remain on the finer side of the observed transition and 40–50 km merge substantially more support into single corridors. This observation explains what the selected

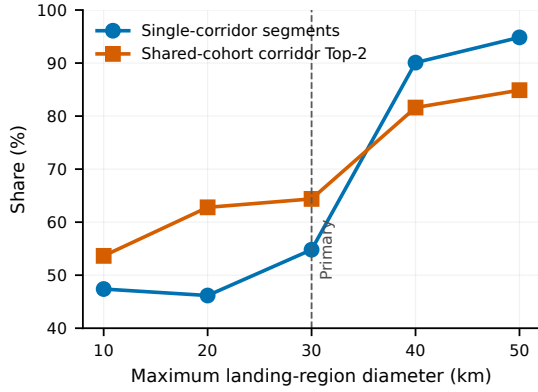


Fig. 6. Recorded A-Root landing-region diameter sensitivity. The analysis does not recompute the main geographic Layer Agreement result.

resolution preserves in that subset; it does not prove that 30 km is parameter-invariant across the corpus.

C. Dependence on Geolocation and Candidate Construction

Every corridor projection begins with hop geolocation. A hop placed farther inland can remove a nearby landing candidate, while a location shifted toward a coast can add one. The 50-km endpoint catchment is the fixed setting in the reported analysis. The current experiment set contains no completed location- perturbation analysis or alternative-geolocation comparison, so the effect of residual IPinfo error is not bounded.

The projection also lacks physical-route ground truth. Geographic, lifecycle, and propagation constraints identify feasible candidates rather than the cable used by a packet. Exact landing-pair, cable, and corridor identities remain linked in the output, permitting later inspection, but the available materials contain no completed manual corridor audit. We therefore report candidate support and avoid statements that require verified cable traversal.

Cable lifecycle filtering uses the inventory fields available for the measurement date. These fields do not report transient failures or outages, and the inventory can be revised. A recorded refresh changed 222 of 694 cable records, including 26 planned-flag changes and 47 revised service dates. The pre-refresh inventory was not used for an existing sensitivity result. Units whose feasible sets lie near a metadata revision may consequently change with the inventory.

D. Snapshot and Population Scope

All 18 measurements cover 00:00–01:00 UTC on July 1, 2026. Routing, service deployment, and cable availability can change outside this window, so the reported distributions describe one aligned snapshot and do not establish temporal stability. A failure-based validation would require comparable traceroutes on both sides of a cable event, which this snapshot does not provide. Subdividing this hour would reuse the same routing interval and could only measure within-window

composition; it would not supply independent evidence of stability across hours or days.

Reported quantities are observation-weighted over participating RIPE Atlas probes. Probe density varies across countries and networks, so a country’s distribution represents the probes and source ASNs present in the corpus, not its user population or traffic volume. The eligibility thresholds of 10 probes and 3 probe ASNs limit sparse units but do not make the sample representative.

The service measurements and dynamic multi-target references also contain different target populations. The references deliberately sample many destinations during the window, whereas each service measurement retains its selected instances. Their differences characterize the observed populations; they are not a matched estimate of a service-selection or deployment effect. Anycast observations jointly reflect deployed footprint, BGP selection, interconnection, and source-network context.

E. Implications for Diversity Measurement

Layer Agreement separates two questions that a network-layer count can otherwise combine. Network Top-2 share records concentration among observed network transitions. Corridor Top-2 share records concentration among feasible physical directions for the same candidate-bearing segments. In this snapshot, the association between the quantities is stronger for island and archipelagic service-facing units than for coastal-mainland units under all three allocation rules.

The existing absolute-breadth results rule out a simple explanation based on island units occupying a universally narrower corridor space. In the DNS country-level summary, island and archipelagic countries have a lower median corridor Top-2 share than other coastal countries (53.3% versus 75.4%) and a higher median effective corridor count (5.58 versus 3.39). Their higher Layer Agreement therefore concerns how network and corridor concentration rankings move together, not uniformly higher corridor concentration.

One mechanism consistent with the measurements is that maritime access and network organization impose partly shared constraints on the two rankings for island and archipelagic sources, whereas terrestrial routing, domestic backhaul, and interconnection can vary network concentration more independently in coastal-mainland settings. The exposure results do not test that mechanism: island and archipelagic countries enter the candidate space more often, but exposure uses all valid traces while Layer Agreement conditions on candidate- bearing segments. Across the 38 countries with both summaries, exposure has associations of -0.29 with corridor Top-2 share and 0.30 with effective corridor count, which do not identify a pathway from geography to agreement.

Other explanations remain compatible with the data, including anycast and service-instance selection, probe and source-AS composition, hop-geolocation error, and geography-dependent candidate ambiguity. The current measurements do not separate these explanations. The geographic result is an association, not evidence that topography, gateway structure,

terrestrial routing, or a particular deployment mechanism causes the difference.

The heterogeneous cross-layer classes also constrain operational conclusions. A broad network distribution projects to a concentrated corridor distribution in 81 service-facing units, but 158 remain broad at both layers and some units expand. A universal conversion from network breadth to physical breadth is therefore unsupported. Exposure and corridor concentration can identify where paths enter the candidate space and how candidate-bearing observations are distributed, but this study does not estimate the effect of deploying a closer service instance, changing interconnection, or adding landing-region access.

Corridor concentration is not an operational resilience metric by itself. It does not measure cable multiplicity within a corridor, landing-station redundancy, shared terrestrial backhaul, correlated hazards, available rerouting capacity, or repair processes. These dimensions are required before a feasible-corridor distribution can support claims about failure tolerance.

F. Ethics and Reproducibility

The study uses archived public RIPE Atlas measurements and infrastructure metadata and reports aggregate country-measurement statistics. It does not target individual users or report individual probe addresses. Candidate assignments are feasible hypotheses and should not be interpreted as disclosure of an operator’s actual cable use.

The released measurement IDs, configuration values, mapping-resolution states, pipeline counts, and scripts reconstruct the primary tables and figures from versioned inputs. The paper’s 80% four-class rule is the primary descriptive classification; the repository’s tier-based auxiliary classification uses a different rule and must remain separately named in the artifact.

VII. CONCLUSION

CLASP compares network-transition and feasible-corridor distributions over the same traceroute segments. In the existing one-hour RIPE Atlas snapshot, projection-score Layer Agreement is 0.632 for 67 island and archipelagic service-facing units and 0.001 for 202 coastal-mainland units; the geographic ordering remains under equal-share and Top-1 allocation. A country-clustered equal-share check retains the ordering but its 95% interval for the 0.461 difference includes zero, so the geographic contrast remains descriptive. The service-facing population is not uniformly contractive: 81 of 275 units are network-broad/corridor-concentrated, while 158 remain broad at both layers. Network-layer diversity therefore does not provide one geography-independent estimate of feasible physical-corridor diversity in the measured populations.

VIII. USE OF AI DISCLOSURE

OpenAI ChatGPT and Codex assisted with code development, plotting scripts and language revision. Their contribution was limited to these tasks, and the authors validated the outputs against the source data and analysis results.

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